

Summer Work at UCSF: Research Abstract

Characterization of Novel Parkinson's Disease Drug Candidates for Target Engagement and Effect on Downstream Gene Expression

An estimated 10 million people worldwide are afflicted by Parkinson's disease, a debilitating neurodegenerative condition not adequately addressed with current therapy. Research has suggested that Parkinson's disease is characterized by low dopamine levels and increased inflammation in the brain.¹ Therefore, upregulating dopamine production and downregulating neuronal inflammation has emerged as a promising therapeutic approach. Dr. Pamela England's lab at UCSF has designed multiple small molecule compounds that bind to the nuclear receptor Nurr1, which controls the expression of a wide range of downstream genes. The small molecules can alter the expression of different downstream genes, depending on their specific structure. One such compound, 87.01, is hypothesized to affect the expression of proteins that produce and package dopamine. Another compound, 85.01, is hypothesized to affect the expression of cytokines and genes encoding for inflammation-related proteins. The precise effect of these small molecules on gene expression, however, remained unclear. Furthermore, evidence supporting the assumption that 87.01 and 85.01 bind to Nurr1, as opposed to some other nuclear receptor, remained elusive.

Throughout the summer, I quantified the gene expression changes induced by 87.01 and two other structural variations, 87-Z-008-G and 87-Z-001-K. In these experiments, I treated cultured cells with varying concentrations of the compounds, and performed RNA extraction and reverse transcription on harvested cell samples. Subsequently, using quantitative polymerase chain reaction (qPCR), I measured the abundance of cellular mRNA for two genes of interest: TH and VMAT2 (both involved in dopamine production).^{2,3,4} We discovered that 87.01 did not induce a statistically significant change in the expression of TH or VMAT2, but interestingly, the two structural analogs caused significant increases in TH and VMAT2 mRNA. In particular, treatment with 87-Z-001-K at 5.0 μ M resulted in a 6.5x increase in TH expression and a 5.2x increase in VMAT2 expression. These data suggest that 87-Z-001-K has the potential to promote the expression of genes associated with dopamine production. Therefore, further development of this compound may hold promise as a treatment for Parkinson's disease.

I performed similar studies on 85.01 and its analogs, and characterized their effects on the expression of genes related to inflammation in microglial cells. In particular, I tracked the expression of three proteins which contribute to the inflammatory response: cytokines IL-1 β and TNF- α , as well as NOS2, an enzyme.⁵ Cells treated with 85.01 demonstrated a decrease in mRNA encoding for all three of these proteins, which highlights 85.01 as a notable compound for further research.

In addition, I investigated whether 87.01 and 85.01 truly bind to Nurr1. This was achieved using a technique known as affinity purification. I chemically modified 87.01 to include a biotin handle. Upon incubation with cells, biotinylated 87.01, in complex with its target receptor, was recovered from cell lysates using plastic beads coated with streptavidin (which has a high affinity for biotin). We then ran the 87.01-receptor complex on a protein gel and used antibodies to detect the presence of Nurr1 or HDAC1—another key regulator of gene expression and a putative

87.01 binding partner. Data collected from two trials suggest that 87.01 may bind to both Nurr1 and HDAC1. Additional studies will be conducted to replicate and confirm these results.

In conclusion, my work on novel Nurr1-targeting compounds suggests they bind to Nurr1, and potentially HDAC, and lead to both the upregulation of genes supporting dopamine production and downregulation of the inflammatory response. Once fully developed, 85.01 and 87.01 used in conjunction with one another may prove effective in producing therapeutic results.

References

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